

Python and R

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This post combines Python and R code demonstrations in Quarto.

Data source: <https://github.com/mcnakhaee/palmerpenguins>

Python Section

```
import pandas as pd
from palmerpenguins import load_penguins
import matplotlib.pyplot as plt
```

```
penguins = load_penguins()
penguins.head()
```

	species	island	bill_length_mm	...	body_mass_g	sex	year
0	Adelie	Torgersen	39.1	...	3750.0	male	2007
1	Adelie	Torgersen	39.5	...	3800.0	female	2007
2	Adelie	Torgersen	40.3	...	3250.0	female	2007
3	Adelie	Torgersen	NaN	...	NaN	NaN	2007
4	Adelie	Torgersen	36.7	...	3450.0	female	2007

[5 rows x 8 columns]

```
df_py = penguins.groupby(['sex', 'island']).agg({'body_mass_g': 'mean'})
ax = df_py.unstack()['body_mass_g'].plot(kind='bar', figsize=(8, 5))
plt.title("Mean Body Mass by Sex and Island (Python)")
plt.ylabel("Mean Body Mass (g)")
plt.xlabel("Sex")
plt.xticks(rotation=0) # Rotate x-axis labels for better readability
```

```
(array([0, 1]), [Text(0, 0, 'female'), Text(1, 0, 'male')])
```

```
plt.legend(title="Island")  
plt.tight_layout()  
plt.show()
```

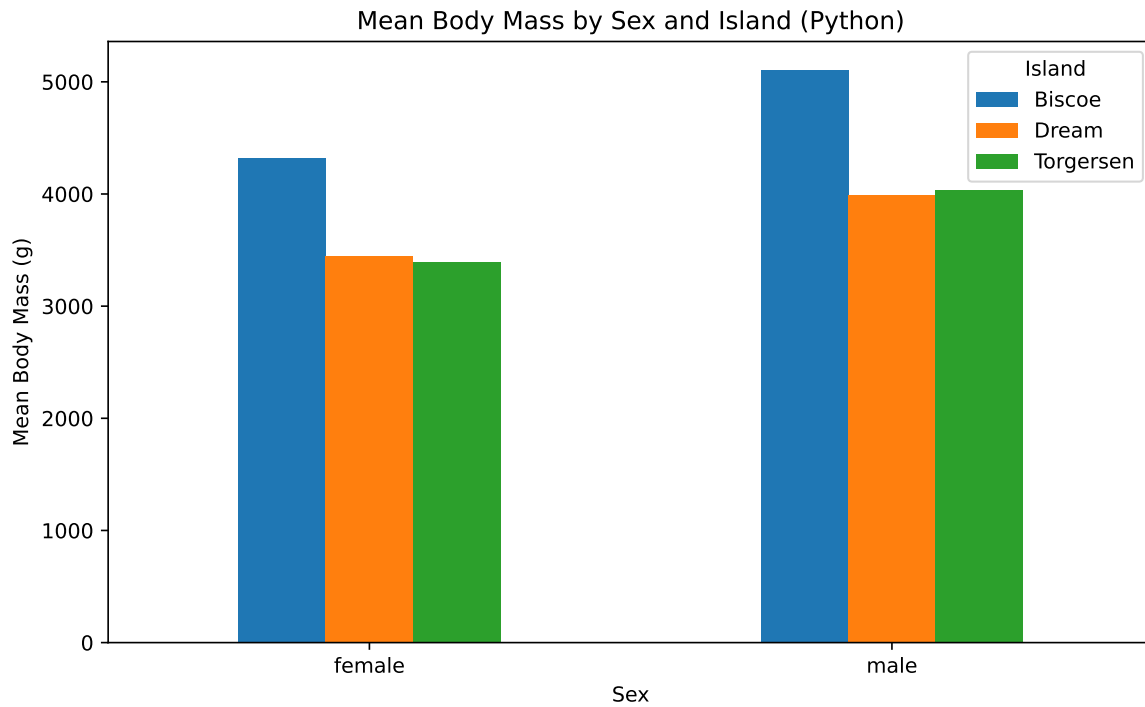


Figure 1: Mean Body Mass by Sex and Island (Python).

R Section

```
#| include: false library(reticulate) use_virtualenv("./.venv", required = TRUE)
```

```
library(dplyr)  
library(ggplot2)  
library(palmerpenguins)  
library(tidyverse)
```

```
data("penguins")  
glimpse(penguins)
```

```

Rows: 344
Columns: 8
$ species      <fct> Adelie, Adelie, Adelie, Adelie, Adelie, Adelie, Adel~
$ island       <fct> Torgersen, Torgersen, Torgersen, Torgersen, Torgerse~
$ bill_length_mm <dbl> 39.1, 39.5, 40.3, NA, 36.7, 39.3, 38.9, 39.2, 34.1, ~
$ bill_depth_mm <dbl> 18.7, 17.4, 18.0, NA, 19.3, 20.6, 17.8, 19.6, 18.1, ~
$ flipper_length_mm <int> 181, 186, 195, NA, 193, 190, 181, 195, 193, 190, 186~
$ body_mass_g   <int> 3750, 3800, 3250, NA, 3450, 3650, 3625, 4675, 3475, ~
$ sex          <fct> male, female, female, NA, female, male, female, male~
$ year         <int> 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007~

```

```

df_r <- penguins |>
  drop_na() |>
  group_by(sex, island) |>
  summarize(mean_mass = mean(body_mass_g, na.rm = TRUE))

```

`summarise()` has regrouped the output.

- i Summaries were computed grouped by sex and island.
- i Output is grouped by sex.
- i Use `summarise(.groups = "drop\_last")` to silence this message.
- i Use `summarise(.by = c(sex, island))` for per-operation grouping (`?dplyr::dplyr_by`) instead.

```

ggplot(df_r, aes(x = sex, y = mean_mass, fill = island)) +
  geom_col(position = "dodge", alpha = 0.85) +
  scale_fill_brewer(palette = "Set2") +
  labs(
    title = "Mean Body Mass by Island & Sex (R)",
    x = "Island",
    y = "Mean Body Mass (g)",
    fill = "Sex"
  ) +
  theme_minimal()

```

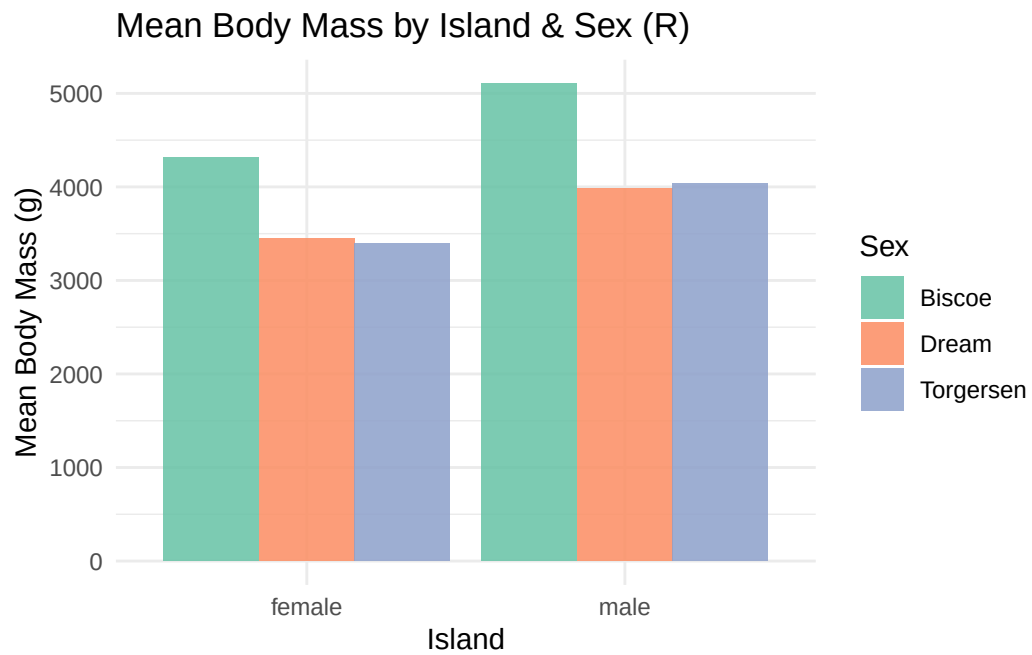


Figure 2: Mean Body Mass by Island & Sex (R).